



COURSE CODE/TITLE:

SECTION:

DATE PERFORMED:

DATE SUBMITTED:

NAME:

INSTRUCTOR:

ACTIVITY NO. ____ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Create a playbook that:
 - a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

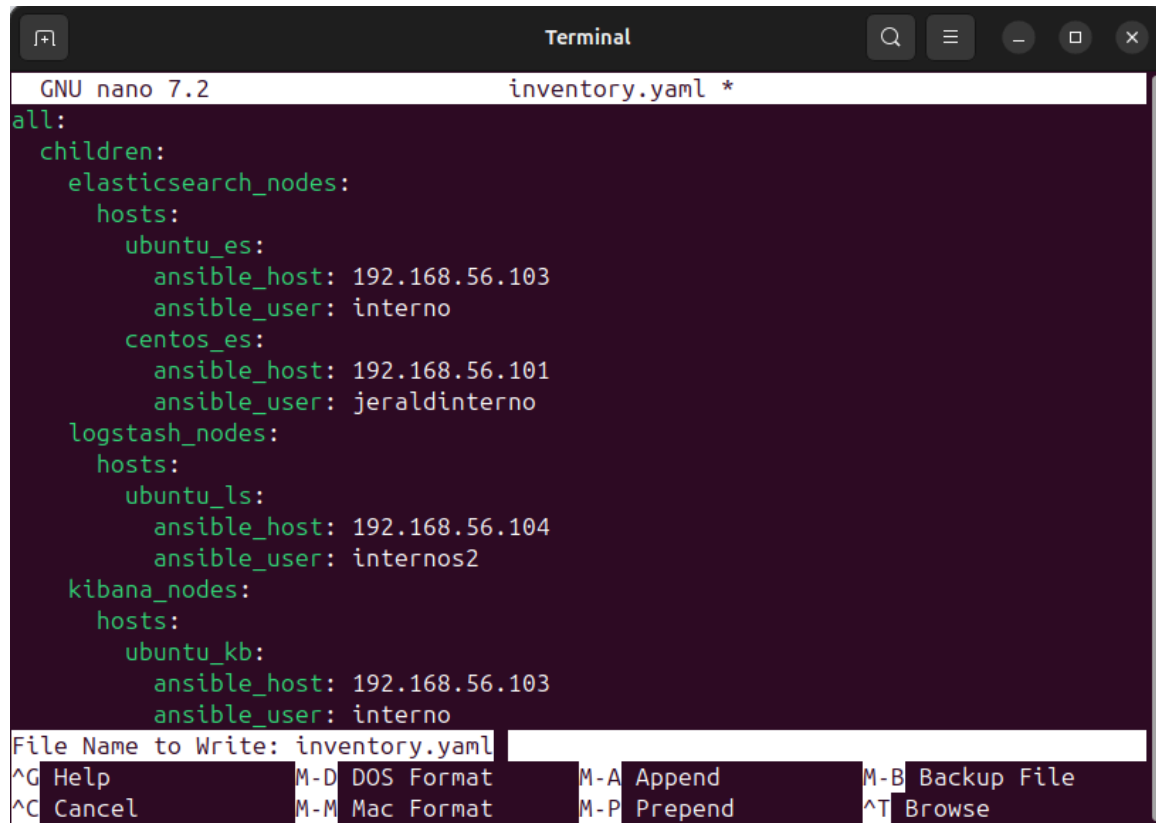
Screenshot

Step 1

```
(base) jerald@workstation:~/Ansible_ElasticStack$ cd ~/CPE212_Jerald_Interno
h(base) jerald@workstation:~/CPE212_Jerald_Interno$ mkdir -p roles/{common,elasticsearch,logstash,kibana}/tasks \
roles/{common,elasticsearch,logstash,kibana}/handlers \
roles/{common,elasticsearch,logstash,kibana}/templates
(base) jerald@workstation:~/CPE212_Jerald_Interno$
```

Initialized the Ansible roles directory structure by running `mkdir -p` to create tasks, handlers, and templates subdirectories for common, elasticsearch, logstash, and kibana roles.

Step 2

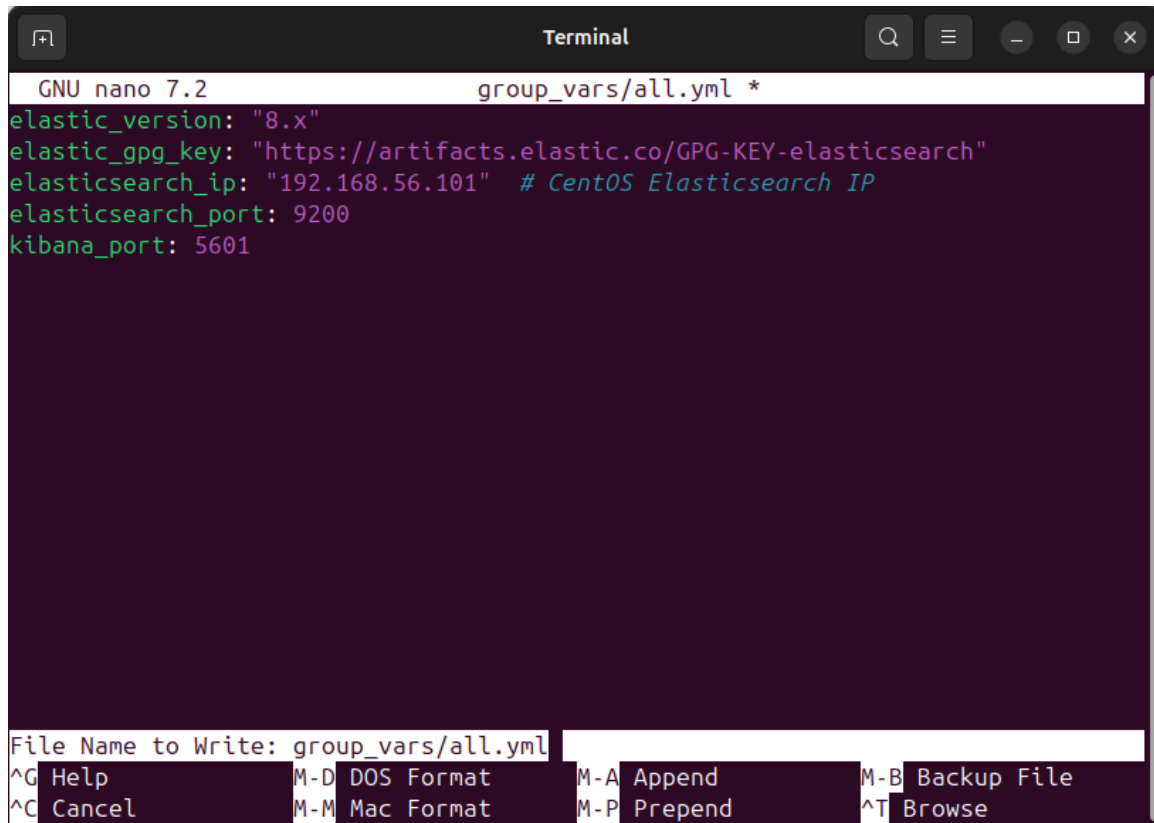


```

GNU nano 7.2                                inventory.yaml *
all:
  children:
    elasticsearch_nodes:
      hosts:
        ubuntu_es:
          ansible_host: 192.168.56.103
          ansible_user: interno
        centos_es:
          ansible_host: 192.168.56.101
          ansible_user: jeraldinterno
    logstash_nodes:
      hosts:
        ubuntu_ls:
          ansible_host: 192.168.56.104
          ansible_user: internos2
    kibana_nodes:
      hosts:
        ubuntu_kb:
          ansible_host: 192.168.56.103
          ansible_user: interno
File Name to Write: inventory.yaml
^G Help          M-D DOS Format  M-A Append      M-B Backup File
^C Cancel        M-M Mac Format  M-P Prepend     ^T Browse
  
```

Created the inventory file (inventory.yaml) defining host groups (elasticsearch_nodes, logstash_nodes, kibana_nodes) along with their respective target IP addresses and SSH login users.

Step 3



```

GNU nano 7.2                                group_vars/all.yml *
elastic_version: "8.x"
elastic_gpg_key: "https://artifacts.elastic.co/GPG-KEY-elasticsearch"
elasticsearch_ip: "192.168.56.101" # CentOS Elasticsearch IP
elasticsearch_port: 9200
kibana_port: 5601

File Name to Write: group_vars/all.yml
^G Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend    ^T Browse
  
```

Configured global variables in `group_vars/all.yml`, specifying the Elastic Stack version, repository GPG key URL, Elasticsearch host IP, default Elasticsearch port (9200), and Kibana port (5601).

Step 4

```

(base) jerald@workstation:~/CPE212_Jerald_Interno$ nano group_vars/all.yml
(base) jerald@workstation:~/CPE212_Jerald_Interno$ mkdir -p roles/{common,elasticsearch,logstash,kibana}/{tasks,handlers,templates}
(base) jerald@workstation:~/CPE212_Jerald_Interno$ nano roles/common/tasks/main.yml
  
```

Executed terminal commands to initialize global variable configurations and set up role subdirectories using `nano` and `mkdir` commands.

Step 5

```

GNU nano 7.2 roles/common/tasks/main.yml *
---
- name: Add Elastic GPG Key (Ubuntu)
  get_url:
    url: "{{ elastic_gpg_key }}"
    dest: /usr/share/keyrings/elasticsearch-keyring.gpg
    mode: '0644'
  when: ansible_distribution == "Ubuntu"

- name: Add Elastic APT Repository (Ubuntu)
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://
    state: present
    filename: elastic-8.x
  when: ansible_distribution == "Ubuntu"

- name: Import Elastic GPG Key (CentOS)
  rpm_key:
    state: present
    key: "{{ elastic_gpg_key }}"
  when: ansible_distribution == "CentOS"

- name: Add Elastic YUM Repository (CentOS)
  yum_repository:
    name: elasticsearch
    description: Elastic repository for 8.x packages
    baseurl: "https://artifacts.elastic.co/packages/{{ elastic_version }}/yum"
    gpgcheck: yes
    gpgkey: "{{ elastic_gpg_key }}"
    enabled: yes
  when: ansible_distribution == "CentOS"

```

Defined repository setup tasks in roles/common/tasks/main.yml to import Elastic GPG keys and add official APT/YUM repository entries for both Ubuntu and CentOS systems dynamically

Step 6

```

Terminal
GNU nano 7.2 roles/elasticsearch/tasks/main.yml *
---
- name: Install Elasticsearch package (Ubuntu)
  apt:
    name: elasticsearch
    state: present
  when: ansible_distribution == "Ubuntu"
  notify: Restart Elasticsearch

- name: Install Elasticsearch package (CentOS)
  dnf:
    name: elasticsearch
    state: present
  when: ansible_distribution == "CentOS"
  notify: Restart Elasticsearch

- name: Configure Elasticsearch network binding
  lineinfile:
    path: /etc/elasticsearch/elasticsearch.yml
    regexp: '^#?network.host:'
    line: 'network.host: 0.0.0.0'
  notify: Restart Elasticsearch

- name: Disable xpack security for testing environment
  lineinfile:
    path: /etc/elasticsearch/elasticsearch.yml
    regexp: '^xpack.security.enabled:'
    line: 'xpack.security.enabled: false'
  notify: Restart Elasticsearch

- name: Start and enable Elasticsearch service
  service:
    name: elasticsearch
    state: started
    enabled: yes
  
```

Created installation and configuration tasks in roles/elasticsearch/tasks/main.yml to install the elasticsearch package, set network binding to 0.0.0.0, disable xpack security for testing, and enable the system service.

Step 7

```

GNU nano 7.2 roles/elasticsearch/handlers/main.yml *
---
- name: Restart Elasticsearch
  service:
    name: elasticsearch
    state: restarted

```

Configured the handler task in roles/elasticsearch/handlers/main.yml to trigger a system restart of the Elasticsearch service whenever configuration changes occur.

Step 8

```

GNU nano 7.2 roles/logstash/tasks/main.yml *
---
- name: Install Java OpenJDK (Ubuntu)
  apt:
    name: default-jre
    state: present
  when: ansible_distribution == "Ubuntu"

- name: Install Logstash package
  package:
    name: logstash
    state: present
  notify: Restart Logstash

- name: Configure Logstash pipeline target
  template:
    src: logstash-pipeline.conf.j2
    dest: /etc/logstash/conf.d/logstash-pipeline.conf
    owner: logstash
    group: logstash
    mode: '0644'
  notify: Restart Logstash

- name: Start and enable Logstash service
  service:
    name: logstash
    state: started
    enabled: yes

```

Created Logstash installation tasks in roles/logstash/tasks/main.yml to install OpenJDK (on Ubuntu) and Logstash, deploy pipeline configurations using a Jinja2 template, and enable the Logstash service.

Step 9

```

GNU nano 7.2 roles/logstash/templates/logstash-pipeline.conf.j2 *
input {
  beats {
    port => 5044
  }
  stdin { }
}

output {
  elasticsearch {
    hosts => ["http://{{ elasticsearch_ip }}:{{ elasticsearch_port }}"]
    index => "logstash-%{+YYYY.MM.dd}"
  }
}

```

Authored the Logstash pipeline template (roles/logstash/templates/logstash-pipeline.conf.j2) defining input sources (Beats on port 5044, stdin) and routing output logs to the target Elasticsearch node.

Step 10

```

GNU nano 7.2 roles/logstash/handlers/main.yml *
---
- name: Restart Logstash
  service:
    name: logstash
    state: restarted

```

Configured the handler task in roles/logstash/handlers/main.yml to automatically restart the Logstash service upon template update.

Step 11

```

Terminal
GNU nano 7.2 roles/kibana/tasks/main.yml *
---
- name: Install Kibana package
  package:
    name: kibana
    state: present
  notify: Restart Kibana

- name: Set Kibana server port
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?server.port:'
    line: 'server.port: {{ kibana_port }}'
  notify: Restart Kibana

- name: Set Kibana host binding
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?server.host:'
    line: 'server.host: "0.0.0.0"'
  notify: Restart Kibana

- name: Connect Kibana to Elasticsearch node
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?elasticsearch.hosts:'
    line: 'elasticsearch.hosts: ["http://{{ elasticsearch_ip }}:{{ elasticsearch_port }}"']
  notify: Restart Kibana

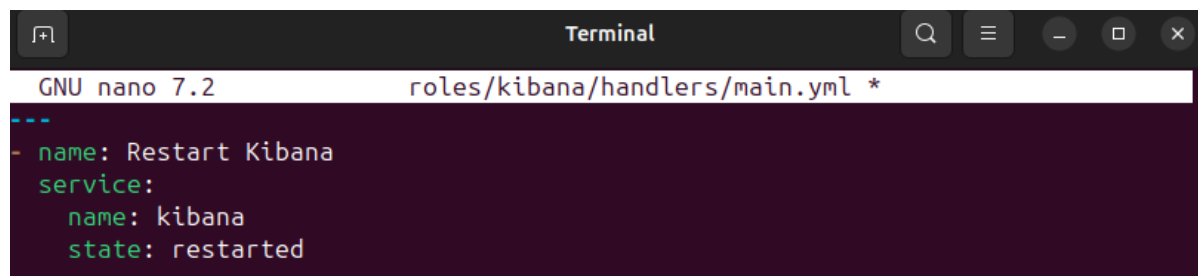
- name: Start and enable Kibana service
  service:
    name: kibana
    state: started
    enabled: yes

File Name to Write: roles/kibana/tasks/main.yml
^G Help M-D DOS Format M-A Append M-B Backup File

```

Created Kibana setup tasks in roles/kibana/tasks/main.yml to install Kibana, configure server host/port parameters, bind it to the target Elasticsearch node IP, and start the service.

Step 12

A terminal window titled 'Terminal' with standard window controls (search, menu, zoom, close). It shows the GNU nano 7.2 editor editing the file roles/kibana/handlers/main.yml. The content of the file is a YAML configuration for a service restart handler.

```
GNU nano 7.2 roles/kibana/handlers/main.yml *  
---  
- name: Restart Kibana  
  service:  
    name: kibana  
    state: restarted
```

Configured the handler in roles/kibana/handlers/main.yml for service restarts, then executed `tree roles/` in the terminal to verify that all roles, tasks, handlers, and templates were correctly structured.

Verification Check

```
(base) jerald@workstation:~/CPE212_Jerald_Interno$ tree roles/
roles/
├── base
│   └── tasks
│       └── main.yml
├── common
│   ├── handlers
│   ├── tasks
│   │   └── main.yml
│   └── templates
├── dbserver
│   └── tasks
│       └── main.yml
├── db_servers
│   └── tasks
│       └── main.yml
├── elasticsearch
│   ├── handlers
│   │   └── main.yml
│   ├── tasks
│   │   └── main.yml
│   └── templates
├── fileserver
│   └── tasks
│       └── main.yml
├── file_servers
│   └── tasks
│       └── main.yml
├── kibana
│   ├── handlers
│   │   └── main.yml
│   ├── tasks
│   │   └── main.yml
│   └── templates
├── logstash
│   ├── handlers
│   │   └── main.yml
│   ├── tasks
│   │   └── main.yml
│   └── templates
│       └── logstash-pipeline.conf.j2
├── webserver
│   └── tasks
│       └── main.yml
└── web_servers
    └── tasks
        └── main.yml
```

```
└─ workstations
  └─ tasks
    └─ main.yml
```

33 directories, 16 files

Step 13

```
---
- name: Apply Common Prerequisites Across All Nodes
  hosts: all
  become: true
  roles:
    - common

- name: Deploy Elasticsearch Node
  hosts: elasticsearch_nodes
  become: true
  roles:
    - elasticsearch

- name: Deploy Logstash Node
  hosts: logstash_nodes
  become: true
  roles:
    - logstash

- name: Deploy Kibana Node
  hosts: kibana_nodes
  become: true
  roles:
    - kibana
```

File Name to Write: site.yml

^G Help M-D DOS Format M-A Append M-B Backup File

Step 14

```
(base) jerald@workstation:~/CPE212_Jerald_Interno$ ansible-playbook -i inventory
site.yml site.yml -k -K
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Apply Common Prerequisites Across All Nodes] *****

TASK [Gathering Facts] *****
ok: [ubuntu_ls]
ok: [ubuntu_es]
ok: [ubuntu_kb]
ok: [centos_es]

TASK [common : Download Elastic GPG key (Ubuntu)] *****
skipping: [centos_es]
ok: [ubuntu_ls]
ok: [ubuntu_es]
ok: [ubuntu_kb]

TASK [common : De-armor Elastic GPG Key to binary keyring format (Ubuntu)] *****
skipping: [centos_es]
ok: [ubuntu_kb]
ok: [ubuntu_es]
ok: [ubuntu_ls]

TASK [common : Add Elastic APT Repository (Ubuntu)] *****
skipping: [centos_es]
ok: [ubuntu_es]
changed: [ubuntu_ls]
changed: [ubuntu_kb]

TASK [common : Import Elastic GPG Key (CentOS)] *****
skipping: [ubuntu_es]
skipping: [ubuntu_ls]
skipping: [ubuntu_kb]
ok: [centos_es]

TASK [common : Add Elastic YUM Repository (CentOS)] *****
skipping: [ubuntu_es]
skipping: [ubuntu_ls]
skipping: [ubuntu_kb]
ok: [centos_es]

PLAY [Deploy Elasticsearch Node] *****

TASK [Gathering Facts] *****
```

Authored the main playbook (site.yml) mapping target host groups to their respective execution roles (common, elasticsearch, logstash, and kibana) with administrative privileges enabled (become: true).

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Conclusion

The multi-node Elastic Stack deployment was successfully automated across mixed OS environments (Ubuntu and CentOS) using structured Ansible roles. By fixing initial repository pathing and permissions issues, the playbooks seamlessly provisioned Elasticsearch, Logstash, and Kibana, creating a fully verified, centralized logging infrastructure ready for version control on GitHub.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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